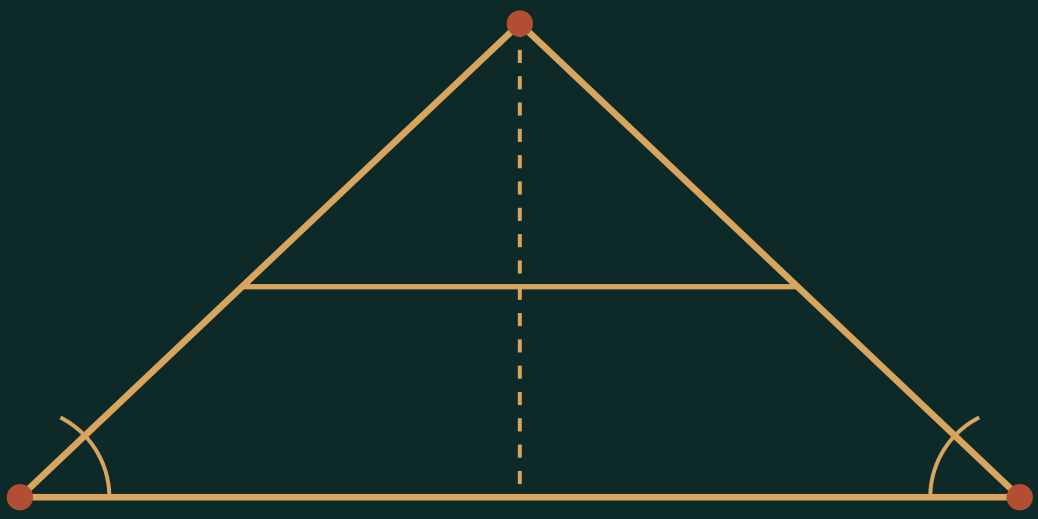
Lines, angles, triangles and constructions

A single straight line contains 180°, and a full turn contains 360°. From those two facts, and one statement about parallel lines, an entire system of shape reasoning follows. Constructions turn that reasoning into a repeatable physical procedure.



equal slopes · horizontal collar · exact template

KEY CONCEPT · RELATIONSHIPS

RELATED · SPACE

RELATED · REPRESENTATION

CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

A workshop needs a triangular roof truss with equal slopes and a horizontal collar beam. Which angles must the carpenter set, how can the shape be checked without measuring every angle, and how can a full-size template be reproduced exactly?

CHAPTER 7 · OPENER

Reasoning with straight lines

Geometry begins with a few reliable facts and becomes powerful when those facts are connected. In this chapter you will name angle relationships, explain triangle and polygon sums, and use equal-radius constructions to force a property rather than guess it.

ENGAGEMENT

Before cutting timber, a carpenter can test a design with a scale drawing. Which measurements are enough to guarantee the shape? Which checks protect against a drawing that only looks correct?

Inquiry questions

- **Factual:** What do angles on a straight line and around a point add to?
- **Conceptual:** Why must the two shorter sides of a triangle together exceed the longest side?
- **Debatable:** When is a construction reliable enough to build from?

Learning goals

- Use angle facts on lines, at points and with parallel lines.
- Reason with triangle and polygon angle sums.
- Apply the triangle inequality and regular-polygon facts.
- Construct perpendicular bisectors, angle bisectors and SSS triangles.

MYP lens

Key concept: Relationships

Related concepts: Space, Representation

Global context: Orientation in space and time

Criteria: A, B, C and D in separate tasks

ATL FOCUS. In order for a student to think creatively and critically, students must choose an appropriate geometric fact or construction, test a design and explain why the method is reliable.

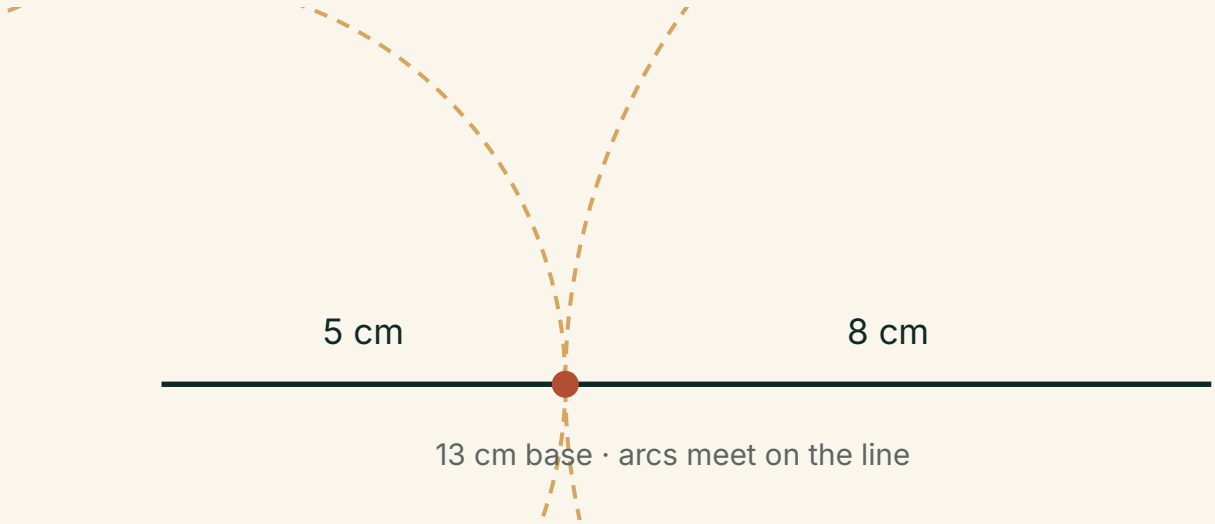
Language bank: acute · right · obtuse · reflex · vertically opposite · corresponding · alternate · co-interior · isosceles · exterior angle · polygon · perpendicular bisector · angle bisector · construction

CHAPTER 7 · INQUIRY LAUNCH

Which lengths close a triangle?

Three strips joined end to end may or may not close a triangle. Predict first, then test with compass arcs. Do not state a rule until the evidence forces it.

SIDE LENGTHS (CM)	PREDICTION	ARCS MEET ABOVE THE BASE?	TRIANGLE?
3, 4, 5			
3, 4, 8			
4, 5, 9			
6, 7, 10			
2, 2, 3			



OBSERVE

When the arcs meet on the base, the “triangle” has zero area. When they do not meet, there is a gap. What must be true about the two shorter lengths for a real apex to exist?

Keep this prediction. You will return to it in the investigation on pages 17–18 and compare your first ideas with the evidence.

CHAPTER 7 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the map from concrete angle facts to constructions and design decisions.

Ready check

Before starting, answer **F1, F3, F6 and F8** from the Foundation bank on page 7. If any answer is uncertain, review the turn model in 7.1 before moving on.

Chapter map

1. **7.1 Lines, angles and angle facts** — name the relationship first: straight line, full turn, vertically opposite or right angle.
2. **7.2 Parallel lines and triangles** — use corresponding, alternate and co-interior angles only when the lines are parallel.
3. **7.3 Polygons and the triangle inequality** — split polygons into triangles; test whether three lengths can close a triangle.
4. **7.4 Ruler-and-compass constructions** — use equal radii to force a property, keep arcs visible and check the defining relationship.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.1 Lines, angles and angle facts

BIG IDEA

Name the relationship first: straight line, full turn, vertically opposite or right angle.

An angle measures a turn. A half-turn along a straight line is 180°, so adjacent angles on a straight line are supplementary. A full turn about a point is 360°, so angles around a point sum to 360°. When two straight lines cross, the angles opposite each other are vertically opposite and equal.

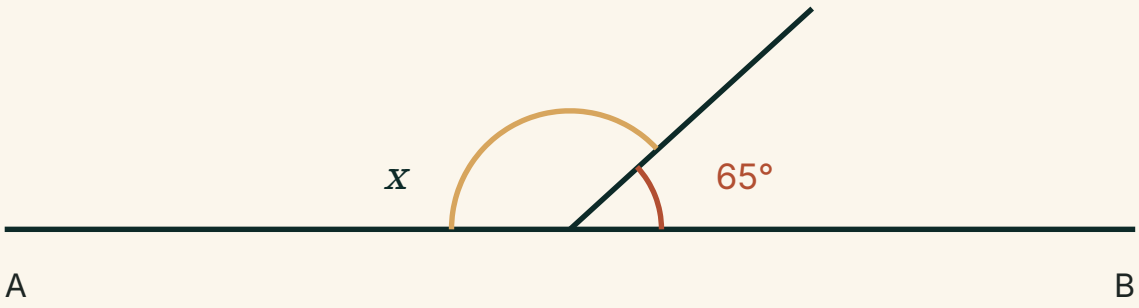


Figure 7.1 · The angle x and 65° lie on a straight line, so x+65°=180°.

Classify angles as acute (<90°), right (90°), obtuse (between 90° and 180°), straight (180°) or reflex (between 180° and 360°). A diagram is not always drawn to scale, so use stated facts rather than appearance.

Key idea

Straight line: 180°. Full turn: 360°. Vertically opposite: equal. Right angle: 90°. The middle letter of ∠ABC names the vertex B.

CHAPTER 7 · WORKED MODELLING

7.1 Worked examples and guided practice

WORKED EXAMPLE 7.1.1 · MISSING ANGLE ON A LINE

Two adjacent angles on a straight line are 47° and x. Find x.

1. Angles on a straight line sum to 180°.
2. x = 180−47 = 133°.
3. Check: 47+133=180.

x=133°.

WORKED EXAMPLE 7.1.2 · ANGLES AROUND A POINT

Four angles meet at a point: 95°, 120°, 85° and x. Find x.

1. Angles around a point sum to 360°.
2. x = 360−(95+120+85) = 60°.
3. Check: 95+120+85+60=360.

x=60°.

COMMON ERROR

Do not swap the totals: a straight line is 180°, a full turn is 360°. Do not treat vertically opposite angles as supplementary; they are equal. The adjacent pair is supplementary.

Guided practice · copy and complete

Angles on a straight line: 118° + ☐ = 180°, so the missing angle is ☐°. Angles at a point: 95° + 120° + 85° + ☐ = 360°, so the missing angle is ☐°. Vertically opposite: if one angle is 47°, the opposite angle is ☐°, and each adjacent angle is 180−☐ = ☐°.

Practice 7.1 · Foundation

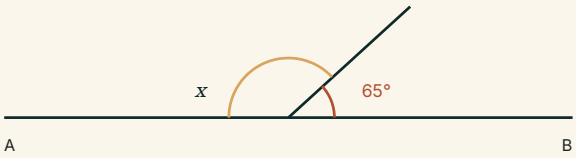
F1–F8

F1 FOUNDATION

Classify each angle: 35°, 90°, 125°, 210°.

F2 FOUNDATION

Find x on the straight line.

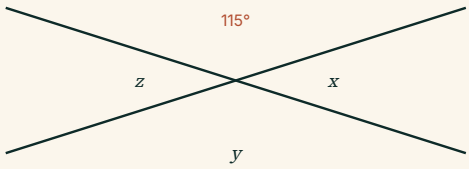


F3 FOUNDATION

Three angles around a point are 90°, 110° and 75°. Find the fourth.

F4 FOUNDATION

Two lines cross. One angle is 115°. Find x, y and z.



F5 FOUNDATION

In a parallel-line diagram, angle 3 and angle 6 are alternate. If angle 3=68°, find angle 6 and the co-interior angle 5.

F6 FOUNDATION

A triangle has angles 35° and 85°. Find the third angle and classify the triangle.

F7 FOUNDATION

Name the polygons with 5, 6 and 8 sides.

F8 FOUNDATION

Which construction locates all points equally distant from A and B? A) angle bisector B) perpendicular bisector C) parallel line D) circle centre A.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.2 Parallel lines and triangles

A transversal is a line that crosses two or more other lines. When the crossed lines are parallel, the eight angles formed fall into equal or supplementary families.

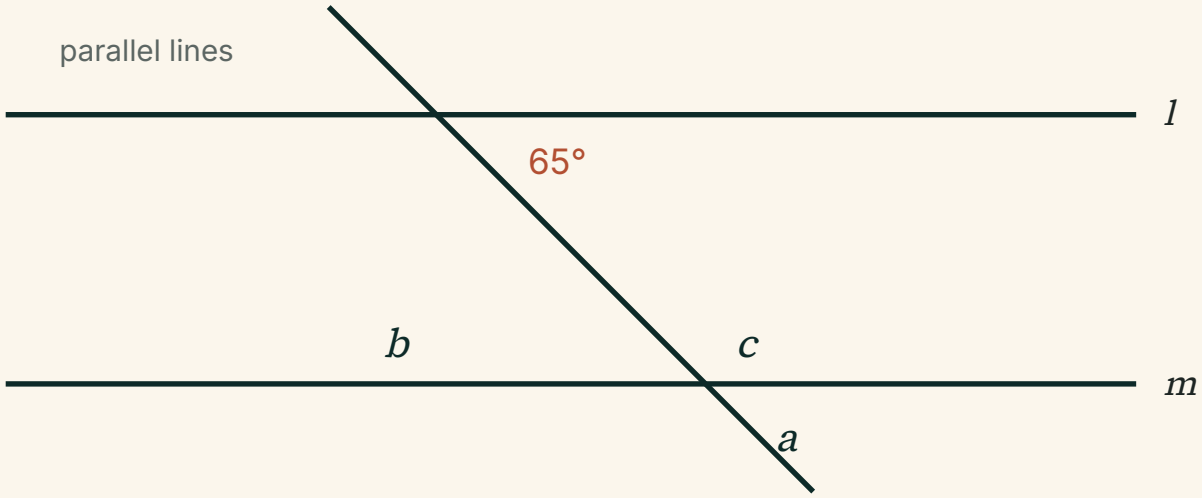


Figure 7.2 · With l ∥ m: corresponding and alternate angles are equal; co-interior angles sum to 180°.

Key idea

Corresponding angles sit in the same position at each intersection and are equal. Alternate angles lie on opposite sides of the transversal between the parallel lines and are equal. Co-interior angles lie on the same side between the parallel lines and are supplementary.

WATCH FOR THIS MISCONCEPTION

Do not assume lines are parallel or angles equal unless marked or stated. “Looks like 90°” is not a reason.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

Triangle sums and exterior angles

Every triangle has an angle sum of 180°. This follows from the parallel-line facts. Triangles are classified by sides as equilateral, isosceles or scalene, and by angles as acute, right or obtuse. In an isosceles triangle the angles opposite the equal sides are equal.

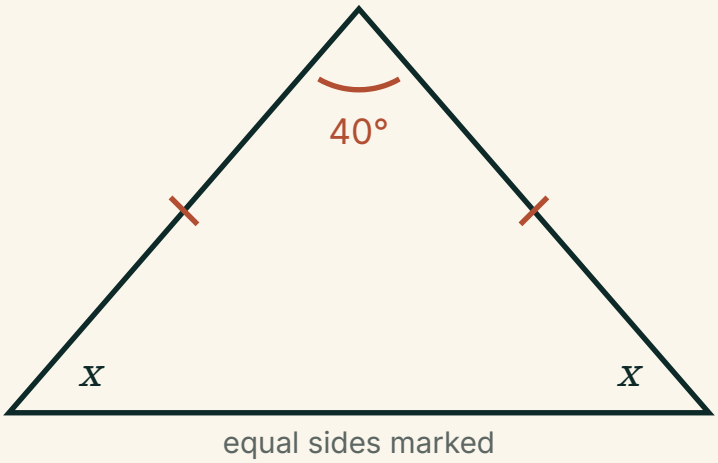


Figure 7.3 · Equal sides force equal base angles.

WORKED EXAMPLE 7.2.1 · ISOSCELES BASE ANGLES

An isosceles triangle has apex angle 40°. Find each base angle.

1. Base angles are equal; call each x.
2. 2x+40=180, so 2x=140.
3. x=70°.

Each base angle is 70°.

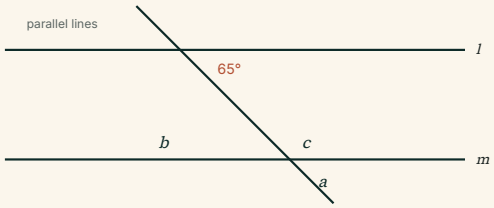
An exterior angle of a triangle equals the sum of the two opposite interior angles. It is supplementary to the interior angle at the same vertex.

Practice 7.2 · Developing

D1–D4

D1 DEVELOPING

Use the parallel-line diagram. Find a, b and c.

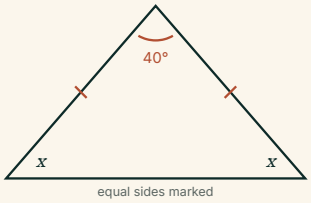


D2 DEVELOPING

The angles of a triangle are x, 2x and 3x. Find x and classify the triangle.

D3 DEVELOPING

Find each base angle x.



D4 DEVELOPING

An exterior angle of a triangle is 115°. One opposite interior angle is 48°. Find the other opposite interior angle and the adjacent interior angle.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.3 Polygons and their angles

A polygon is a closed plane shape made from straight sides. Choose one vertex and draw diagonals to every non-adjacent vertex. The polygon is divided into triangles. An n-sided polygon divides into n−2 triangles, so its interior angle sum is (n−2)×180°.

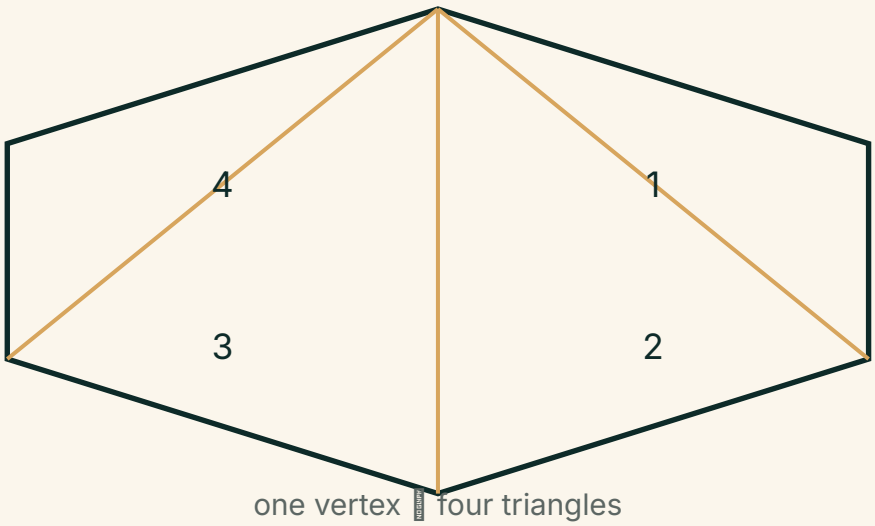


Figure 7.4 · A hexagon splits into four triangles; the interior angle sum is 4×180°=720°.

WORKED EXAMPLE 7.3.1 · REGULAR POLYGON

A regular polygon has exterior angle 45°. Find the number of sides and each interior angle.

1. Exterior angles of any convex polygon sum to 360°.
2. n = 360÷45 = 8.
3. Interior = 180−45 = 135°.

8 sides; each interior angle 135°.

Key idea

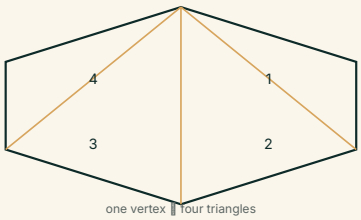
Interior sum: (n−2)×180°. Exterior sum: 360°. Regular polygon: each exterior = 360÷n; each interior = 180−360÷n.

Practice 7.3 · Developing

D5–D8

D5 DEVELOPING

Complete the table: quadrilateral ☐ __ triangles, __°; pentagon ☐ __ triangles, __°; hexagon ☐ __ triangles, __°.

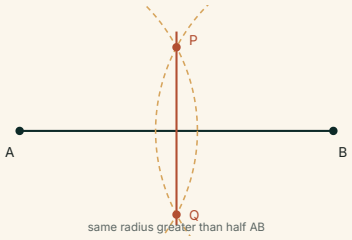


D6 DEVELOPING

A regular polygon has exterior angle 45°. Find the number of sides and each interior angle.

D7 DEVELOPING

Order the steps for a perpendicular bisector of AB=8 cm: (i) draw arcs from A and B with the same radius; (ii) draw AB; (iii) join the arc intersections; (iv) open the compass to more than half of AB. State the radius condition.



D8 DEVELOPING

Test whether these triples can form a triangle: 3,4,8; 5,5,9; 6,7,10.

Guided practice · copy and complete

A pentagon has ☐ sides, splits into ☐ triangles, so its interior angle sum is ☐×180° = ☐°. A regular hexagon has each exterior angle 360÷☐ = ☐°, so each interior angle is 180−☐ = ☐°.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.4 Constructions with ruler and compass

A construction is an argument made with instruments. Each arc records a fixed distance from a named point, so the finished drawing contains evidence: equal radii, equal distances, and a property that can be checked.

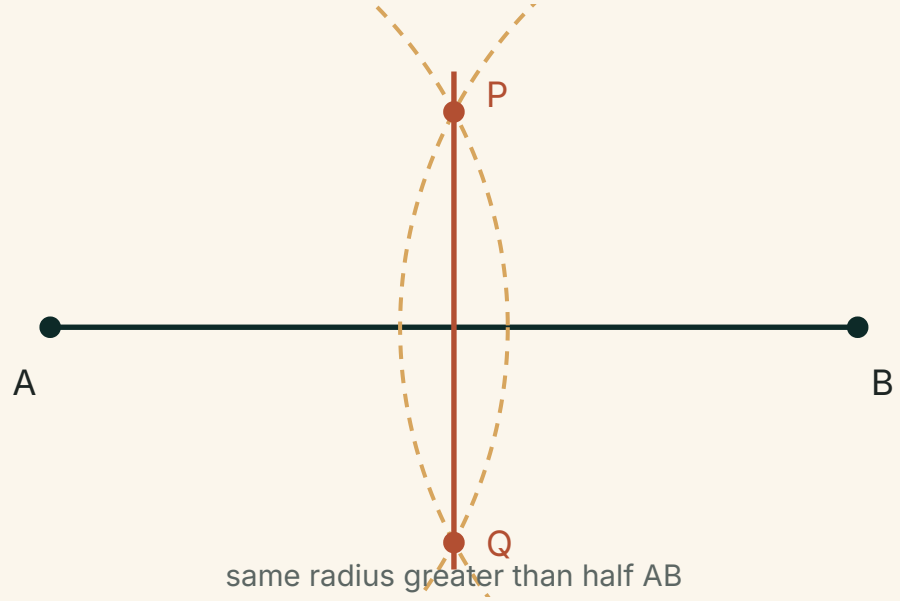


Figure 7.5 · The perpendicular bisector of AB is the line through the two equal-radius arc intersections.

CONSTRUCTION	COMPASS SETTING	HOW TO TEST
Perpendicular bisector of AB	One radius greater than half AB, unchanged from A and B	Crosses AB at its midpoint and makes 90°
Angle bisector	Arc from vertex cuts both arms; same radius from those points	Two half-angles are equal
Triangle from three sides	Base equal to one side; radii equal to the other two sides	Arcs meet; two shorter sides exceed the longest

CHAPTER 7 · WORKED MODELLING

7.4 Worked construction and guided steps

WORKED EXAMPLE 7.4.1 · SSS TRIANGLE

Construct a triangle with sides 5 cm, 6 cm and 7 cm.

1. Draw base AB=7 cm.
2. Open the compass to 5 cm; draw an arc from A.
3. Open the compass to 6 cm; draw an arc from B.
4. Label the intersection C and join AC and BC.
5. Check: 5+6>7, so the arcs meet above the base.

Triangle ABC has the required three sides. Keep the arcs visible.

COMMON ERROR

Never change the compass between a pair of arcs. If the radius is not greater than half the segment, the arcs for a perpendicular bisector will not meet. Check the defining property, not the appearance of the drawing.

Guided practice · order the steps

Put these steps for a perpendicular bisector in order: (i) keep the same radius and draw arcs from A and B; (ii) draw segment AB; (iii) draw the line through the two intersection points; (iv) open the compasses to more than half of AB. Correct order: __, __, __, __. The line is perpendicular because both intersection points are the same distance from __ and from __.

Application · The roof-truss workshop

A1–A4

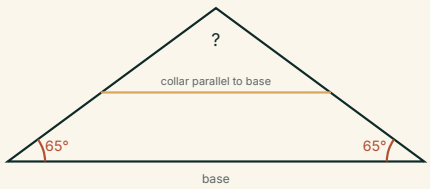
A carpenter is making a triangular roof truss for a garden shelter. Use angle facts, scale and constructions to check the design before cutting timber.

A1 APPLICATION

A ramp makes 12° with the ground. What angle does it make with a vertical post?

A2 APPLICATION

The truss has equal base angles of 65°. Find the apex angle, the exterior angle at a base, and classify the triangle.

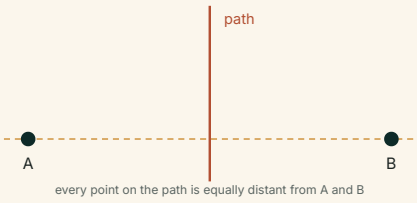


A3 APPLICATION

A scale drawing uses 1:20. The model base is 30 cm. Find the real base in metres.

A4 APPLICATION

A path must pass through points equally distant from gates A and B. Which construction locates the path? Explain.



Application · Collar, kite and error check

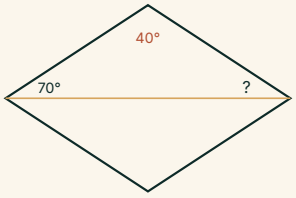
A5–A7

A5
APPLICATION

A collar is parallel to the base. At a rafter joint it makes a 66° co-interior angle. Find the angle on the other side of the collar at that joint.

A6
APPLICATION

The kite is made from two congruent triangles. One triangle has angles 40° and 70°. Find the missing angle and explain why the matching angle in the other triangle is equal.



two congruent triangles

A7
APPLICATION

A student says a triangle with sides 4 cm, 4 cm and 9 cm is possible because 4+9>4. Correct the reasoning.

EVALUATE

Which check protects the design most: measuring one angle, checking the triangle inequality, or keeping construction arcs visible? Explain your choice.

CHAPTER 7 · INVESTIGATION · CRITERION B

The triangle inequality

Return to the prediction on page 3. Choose a longest side c and two shorter sides a and b. Set the base to c, then draw an arc of radius a from one endpoint and an arc of radius b from the other. A triangle exists exactly when the two arcs meet above the base.

SIDES A, B, C (C LONGEST)	3,4,5	3,4,8	4,5,9	6,7,10	2,2,3
a+b	7	7	9	13	4
Triangle possible?					

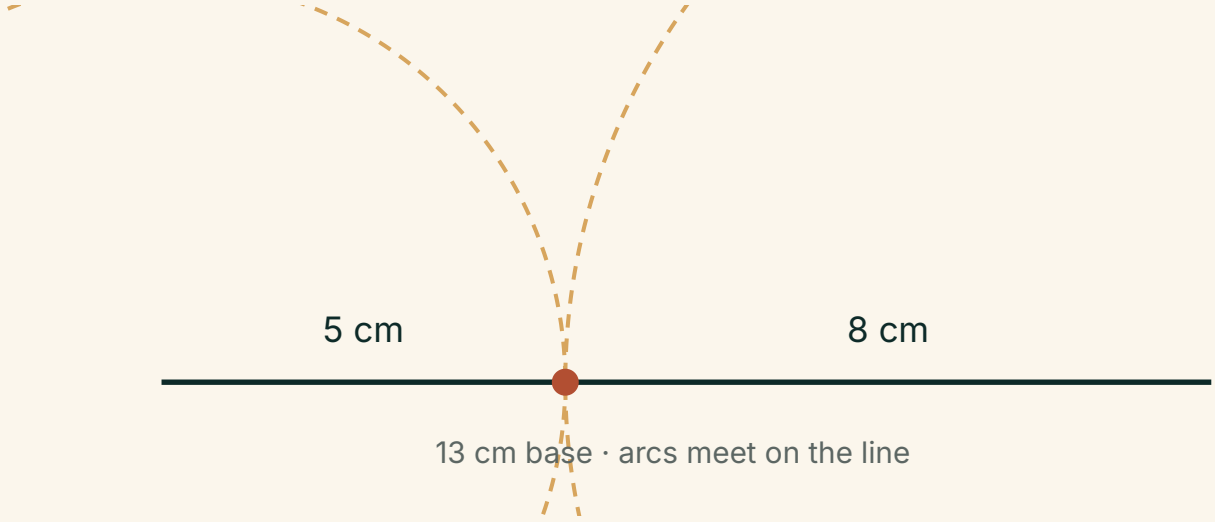


Figure 7.6 · Arcs of radius 5 cm and 8 cm from the ends of a 13 cm base meet on the base, not above it.

1. Complete the table. Explain what happens when the strips 4, 5 and 9 are joined.
2. Compare a+b with c for each row. Write a rule that predicts whether a triangle can be made.
3. Test the rule on 5,8,13; 5,8,12; and 7,7,14. Which cases are borderline?

CHAPTER 7 · INVESTIGATION · DISCOVERY AND CONCLUSION

From arcs to a rule

DISCOVERY

The two shorter sides must together be **strictly greater** than the longest side. If a+b=c, the arcs meet on the base and the triangle has zero area. If a+b<c, the arcs do not meet.

Conclusion

For sides a, b, c with c longest, a triangle exists if and only if a+b>c. The other two inequalities then hold automatically. This is the triangle inequality.

R1 REASONING

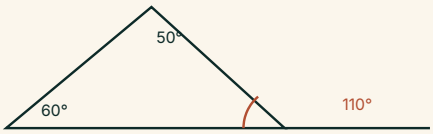
Can a triangle have two obtuse angles? Explain.

R2 REASONING

True or false: any two angles between parallel lines are equal. Justify.

R3 REASONING

A student says the exterior angle 110° should be added to the opposite interior angles 60° and 50° to make 180°. Identify the error and correct it.



exterior angle = sum of the two opposite interior angles

Reasoning · Explain and justify

R4–R7

R4 REASONING

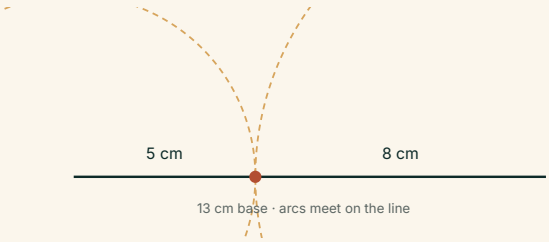
Complete the missing steps: a heptagon has __ sides, splits into __ triangles, so its interior angle sum is __.

R5 REASONING

A student uses a compass radius of 2 cm to bisect a 6 cm segment. The arcs do not meet. Explain why and state a suitable radius.

R6 REASONING

Use the arcs to explain why sides 5 cm, 8 cm and 13 cm do not form a triangle.



R7 REASONING

Why should construction arcs be kept visible?

Challenge · Deeper thinking

C1–C5

C1 CHALLENGE

Two sides of a triangle are 7 cm and 11 cm. The third side is a whole number of centimetres. Find all possible values.

C2 CHALLENGE

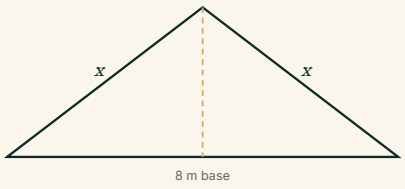
Construct an isosceles triangle with equal sides 6 cm and base 8 cm. Explain why the perpendicular bisector of the base passes through the apex.

C3 CHALLENGE

A regular polygon has each interior angle five times its exterior angle. Find the number of sides.

C4 CHALLENGE

A truss has an 8 m base and equal rafters of length x m. Find the smallest whole-number value of x that makes a triangle.



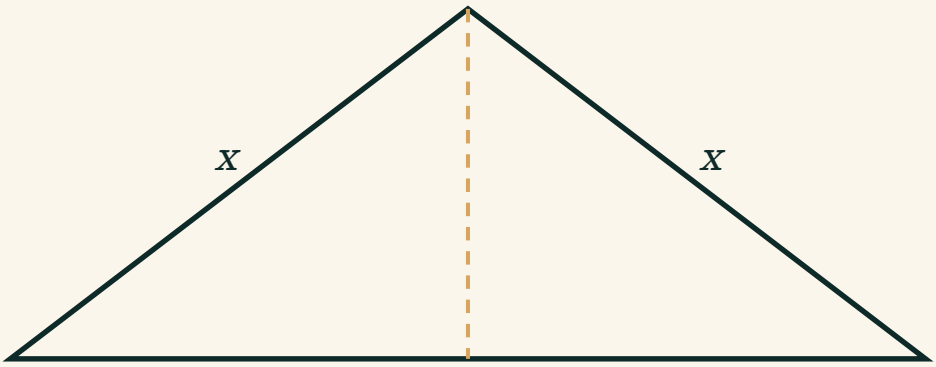
C5 CHALLENGE

Use compass arcs to explain why the two shorter sides of a triangle must together exceed the longest side.

CHAPTER 7 · ASSESSMENT PRACTICE

The truss card

Write a step-by-step card for a partner who must construct an isosceles truss with an 8 m base and equal rafters. The card must be usable with a ruler and compasses, not a protractor.



Criteria A and C

Choose a whole-number rafter length that makes a triangle. Explain why it works, give the construction steps, and state how your partner can check the result without relying on appearance.

Criterion D

Evaluate the design: what happens if the rafter length is exactly 4 m? What safety margin would you recommend, and why? Use the triangle inequality and the construction evidence in your answer.

EVIDENCE

Your card should show: a valid length (A), a clear construction and check (C), and an evaluation of the borderline case in context (D). Do not claim Criterion B unless you include an investigation.

CHAPTER 7 · MIXED PRACTICE

Connect ideas across the chapter

Do not use a calculator. Draw a sketch, mark the known angles and give a reason for each step. This route revisits items from the bank in a mixed order.

ROUTE	ITEMS	FOCUS
Fluency	F2, F3, F4, D6	Angle facts and regular polygons
Visual reasoning	D1, D3, D5, A2	Diagrams, triangles and trusses
Construction	D7, A4, R5, R7	Equal radii and evidence
Inquiry and challenge	R6, C1, C4, C5	Triangle inequality and design

CHECK YOUR PROGRESS

Match the fact to the position: a line gives 180°, a point gives 360°, and co-interior angles are supplementary, not equal. Use the correct pair: alternate and corresponding angles are equal only when the crossed lines are parallel.

CHAPTER 7 · REVIEW

Summary and connections

Angle facts

- Straight line: 180°.
- Full turn: 360°.
- Vertically opposite: equal.
- Parallel lines: corresponding and alternate equal; co-interior supplementary.

Triangles and polygons

- Triangle sum: 180°.
- Exterior angle: sum of opposite interior angles.
- n-gon interior sum: (n−2)×180°.
- Exterior sum: 360°.

Constructions

- Perpendicular bisector: equal radii greater than half the segment.
- Angle bisector: equal arcs from the arms.
- SSS triangle: arcs meet when the triangle inequality holds.

Inquiry

- Triangle inequality: two shorter sides > longest.
- Equality gives a degenerate straight line.
- Keep arcs visible as evidence.

CONNECTIONS · DRAWING IS NOT PROVING

A construction is an argument made with instruments. Each arc records a fixed distance from a named point, so the finished drawing already contains evidence: equal radii, equal distances, and a property that can be checked. A picture that looks right is not enough; the defining relationship must hold.

Where this leads: Chapter 8 turns classification into a precise language. Sets describe which objects share a property, and Venn diagrams show relationships of overlap and exclusion — the same kind of structural thinking used to sort triangles and angles here. Chapter 9 then measures the shapes whose boundaries you have learned to reason about.

CHAPTER 7 · CHECK YOUR PROGRESS

Reflection and next step

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: ITEM OR TASK	NEXT STEP
I can name and classify angles.			
I can use angle facts with reasons.			
I can calculate triangle and polygon angles.			
I can carry out and explain constructions.			
I can decide whether three lengths form a triangle.			
I can produce and evaluate a scale drawing.			

REFLECTION

Which inquiry question changed your thinking most: the factual question about angle totals, the conceptual question about why the inequality must hold, or the debatable question about when a construction is reliable enough to build from? Use one item from the bank as evidence.

ATL EVIDENCE. In order for a student to think creatively and critically, students must choose an appropriate geometric fact or construction, test a design and explain why the method is reliable. Point to the place in your work where you did this.

Review the answer key only after attempting each question. A correct answer without a reason is only half an explanation.

CHAPTER 7 · ANSWERS AND REASONING

Answers · Foundation and Developing

F1 Acute, right, obtuse, reflex.
Compare with 90°, 180° and 360°.

F2 115°.
Angles on a straight line sum to 180°: 180−65.

F3 85°.
360−(90+110+75).

F4 x=65°, y=115°, z=65°.
Vertically opposite angles are equal; adjacent angles sum to 180°.

F5 Angle 6=68°; angle 5=112°.
Alternate angles are equal; co-interior angles sum to 180°.

F6 60°; acute scalene.
180−35−85; all angles are acute and different.

F7 Pentagon, hexagon, octagon.
Polygon names count sides.

F8 B.
The perpendicular bisector is the locus of points equidistant from A and B.

D1 a=65°, b=65°, c=115°.
Corresponding and alternate angles are equal; co-interior angles sum to 180°.

CHAPTER 7 · ANSWERS AND REASONING

Answers · Developing and Application

D2 x=30°; angles 30°, 60°, 90°; right triangle.
x+2x+3x=180.

D3 70° each.
(180−40)÷2; base angles of an isosceles triangle are equal.

D4 67° and 65°.
115−48; 180−115.

D5 Quadrilateral: 2 triangles, 360°. Pentagon: 3, 540°. Hexagon: 4, 720°.
An n-gon splits into n−2 triangles; multiply by 180°.

D6 8 sides; interior 135°.
360÷45; 180−45.

D7 ii, iv, i, iii; radius >4 cm.
The arcs must meet; the line through the intersections is perpendicular and bisects AB.

D8 No; yes; yes.
Compare the sum of the two shorter sides with the longest.

A1 78°.
Ground and vertical are perpendicular: 90−12.

A2 Apex 50°; exterior 115°; isosceles.
180−65−65; 180−65; equal base angles imply equal rafters.

CHAPTER 7 · ANSWERS AND REASONING

Answers · Application and Reasoning

A3 6 m.
30×20=600 cm=6 m.

A4 Perpendicular bisector of AB.
Every point on the perpendicular bisector is equidistant from A and B.

A5 114°.
Co-interior angles sum to 180°.

A6 70°; corresponding angles of congruent triangles are equal.
180−40−70; congruence preserves angles.

A7 Impossible: the two shorter sides give 4+4<9.
The triangle inequality must be checked using the two shorter sides against the longest.

R1 No.
Two obtuse angles already sum to more than 180°.

R2 False.
Corresponding and alternate angles are equal; co-interior angles are supplementary.

R3 The exterior angle equals the sum of the two opposite interior angles: 110°=60°+50°. It is not added to them to make 180°.
The adjacent interior angle is 70°; 110° and 70° are supplementary.

R4 7 sides; 5 triangles; 900°.
(7−2)×180.

CHAPTER 7 · ANSWERS AND REASONING

Answers · Reasoning and Challenge

R5 The radius must be greater than half of 6 cm, so greater than 3 cm; 2 cm is too small.
Equal-radius arcs meet only when the radius exceeds half the segment.

R6 They form a degenerate straight line because 5+8=13.
Equality gives zero area; the two shorter sides must exceed the longest.

R7 They record the equal radii and the defining property, so the construction can be checked.
A construction is an argument made with instruments.

C1 5 through 17 cm.
$11-7 < x < 11+7$, so $4 < x < 18$.

C2 The apex is equidistant from the base endpoints, so it lies on the perpendicular bisector.
Equal sides place the apex on the locus of points equidistant from the endpoints.

C3 12 sides.
Interior + exterior $=180°$; $6e=180°$, $e=30°$, $n=360÷30$.

C4 5 m.
$2x>8$, so $x>4$; $x=4$ gives a degenerate straight line.

C5 If the base is the longest side c, arcs of radius a and b from the endpoints meet above the base only when a+b>c. If a+b=c they meet on the base; if a+b
The arc intersection is the possible apex; no intersection means no triangle.